3D GIS: Working with the 3D Object Layer

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ESRI EUROPEAN DEVELOPER & TECHNOLOGY SUMMIT 2026

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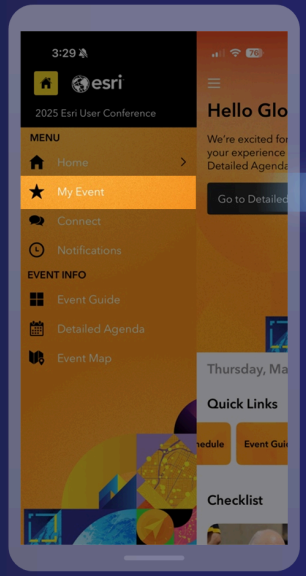

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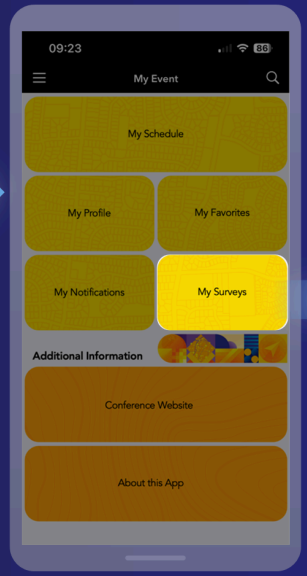

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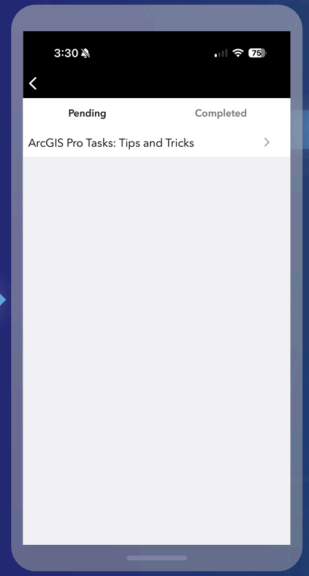

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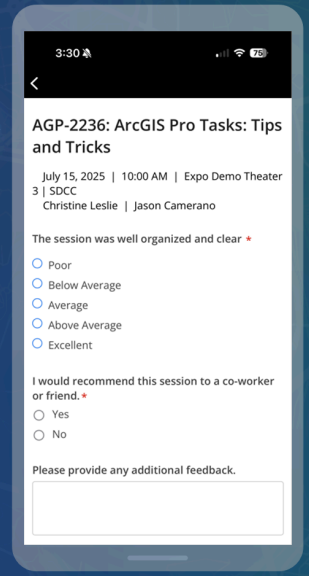


OUR ROUTE

From a model file to a GIS workflow

01

Understand

Models, features and the scene cache

02

Create & upload

Publish a layer and place a model

03

Locate correctly

Origins, vertices and spatial references

04

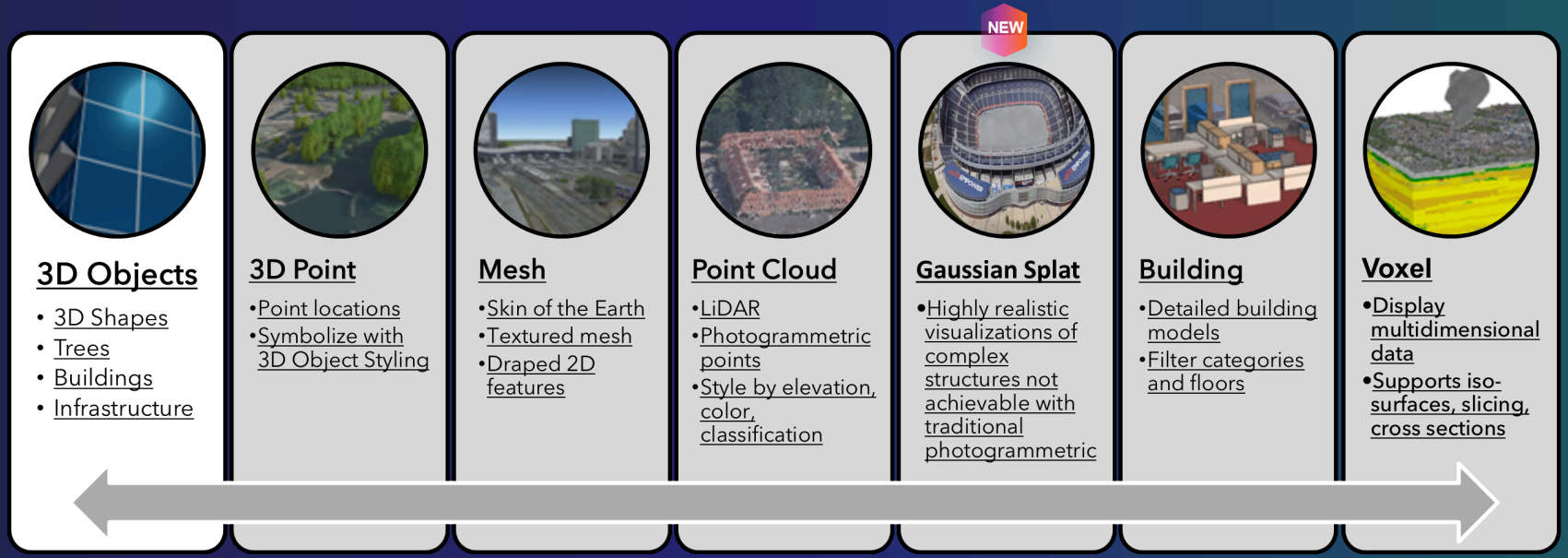
Build on it

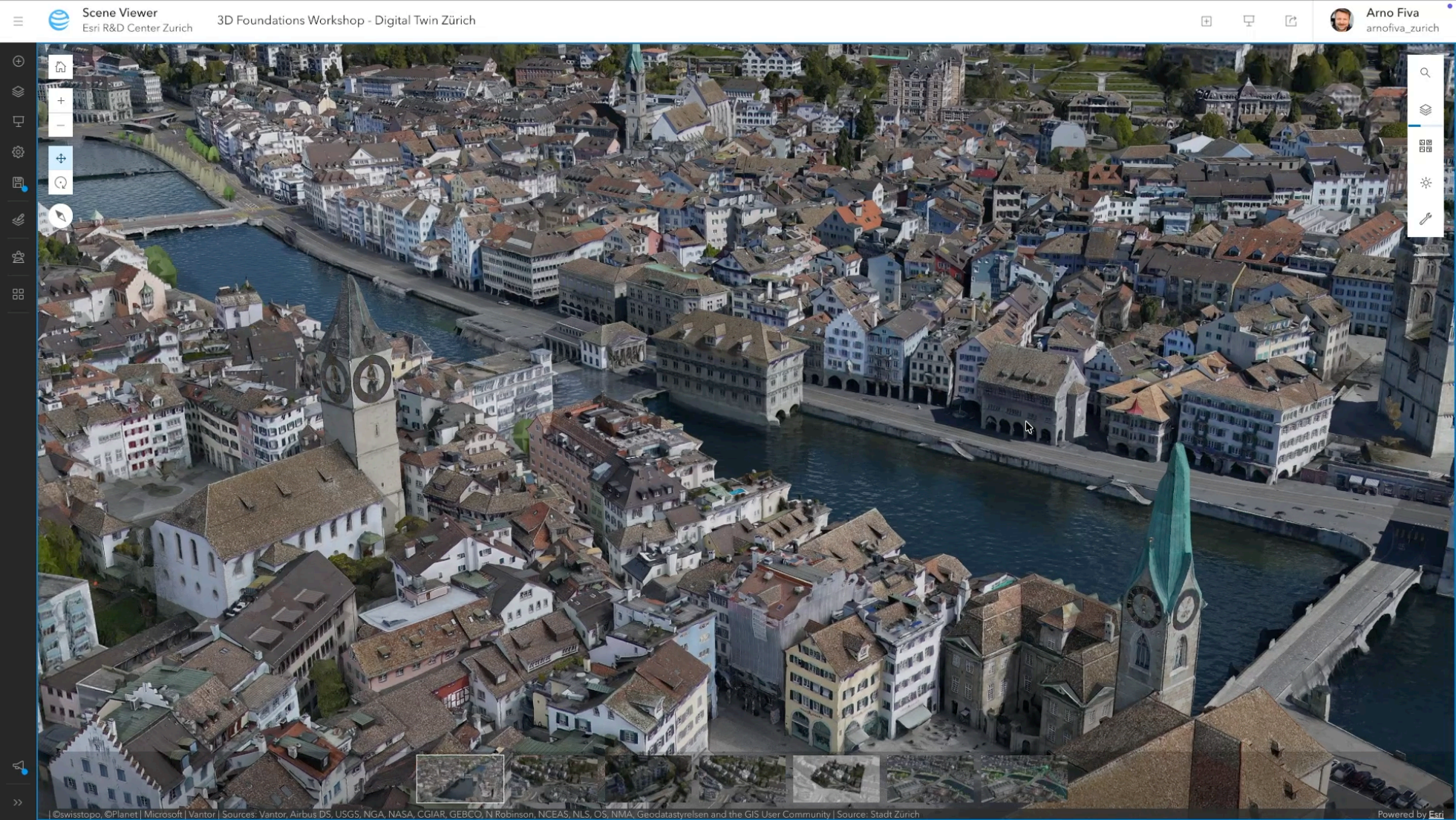
Query, analyze and export

3D Object Layer key concepts

3D LAYER TYPES · SESSION SCOPE

Our focus: 3D object layers for buildings




Scene Viewer
Esri R&D Center Zurich
3D Foundations Workshop - Digital Twin Zürich
Arno Fiva
arnofiva_zurich
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ocolats Lindt

KEY CONCEPTS

A 3D model becomes an editable GIS feature

IFC · FBX
OBJ · GLB

Original model

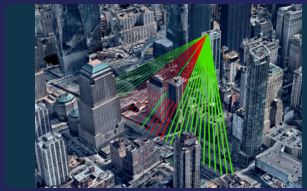
→

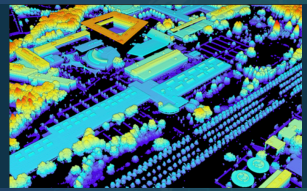
3D OBJECT LAYER

Location + attributes + model

Retain the original asset.
Generate representations for web display.

> A system of record for geolocated 3D objects.

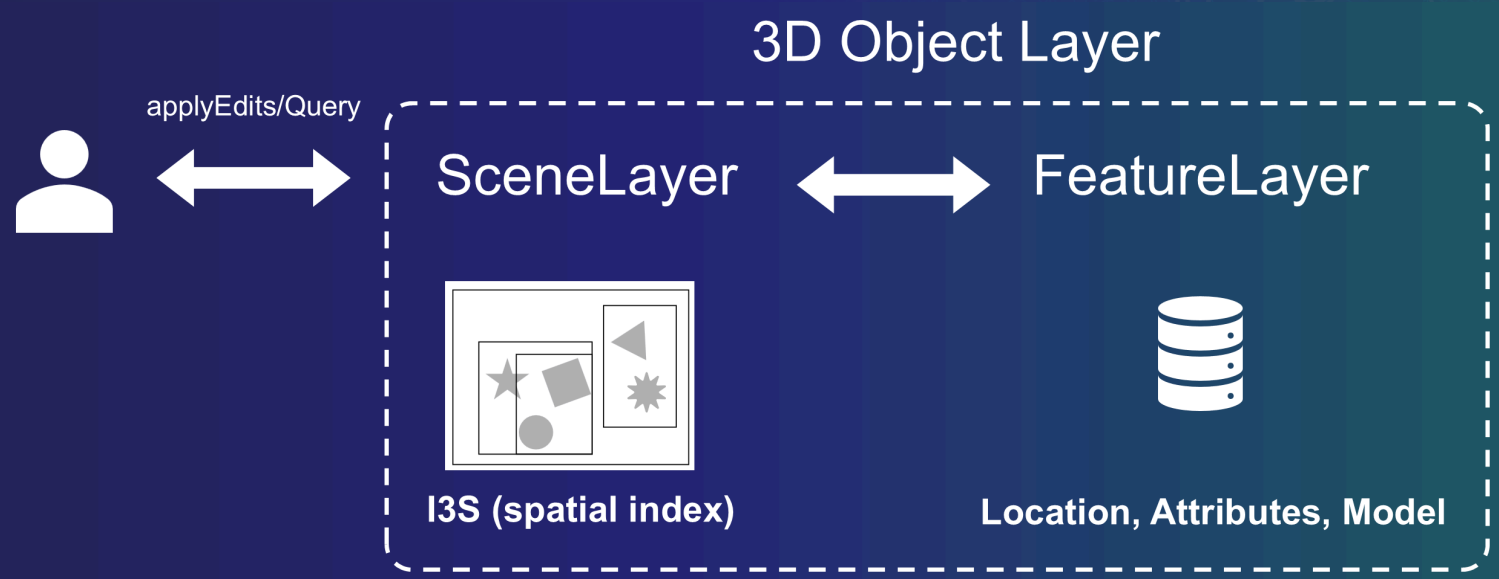




Scene layers and associated feature layers

TWO PARTS, ONE LAYER

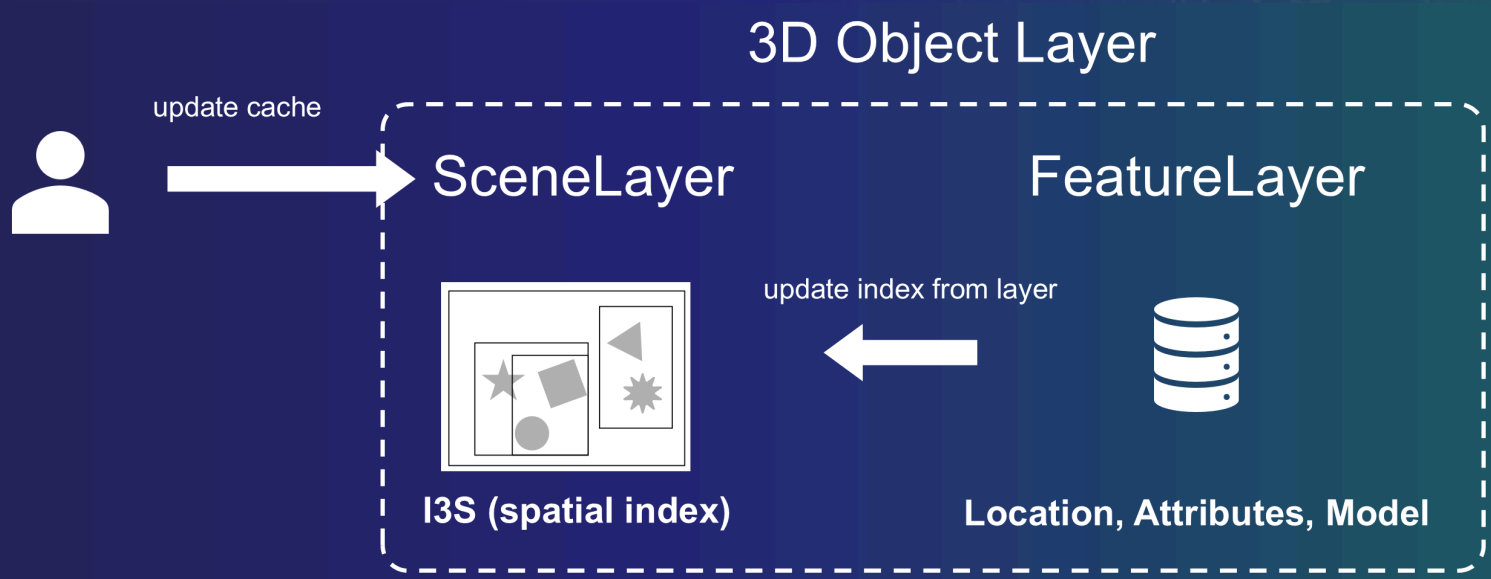
The scene layer displays; the feature layer stores



Use the SceneLayer API for the standard SDK workflow.

CACHING

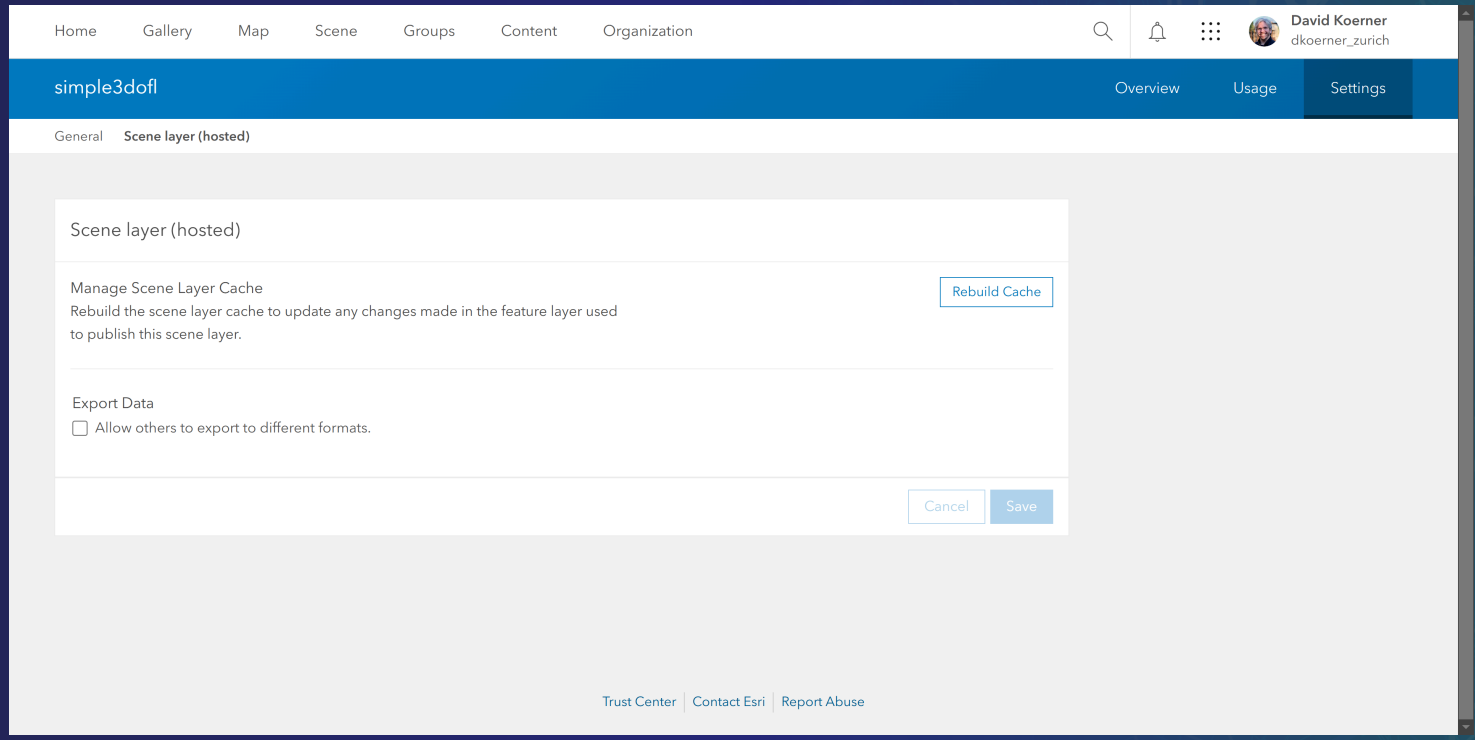
Edits appear before the cache is rebuilt



More uncached edits can mean more work for the browser.

MANAGE THE CACHE

Rebuild after a batch of changes



THE DEVELOPER'S VIEW

One layer, two services, familiar APIs

Scene layer

Fast display
I3S cache and levels of detail

Feature layer

Spatial database
Location, attributes and original models

SceneLayer API

`queryFeatures()`

`applyEdits()`

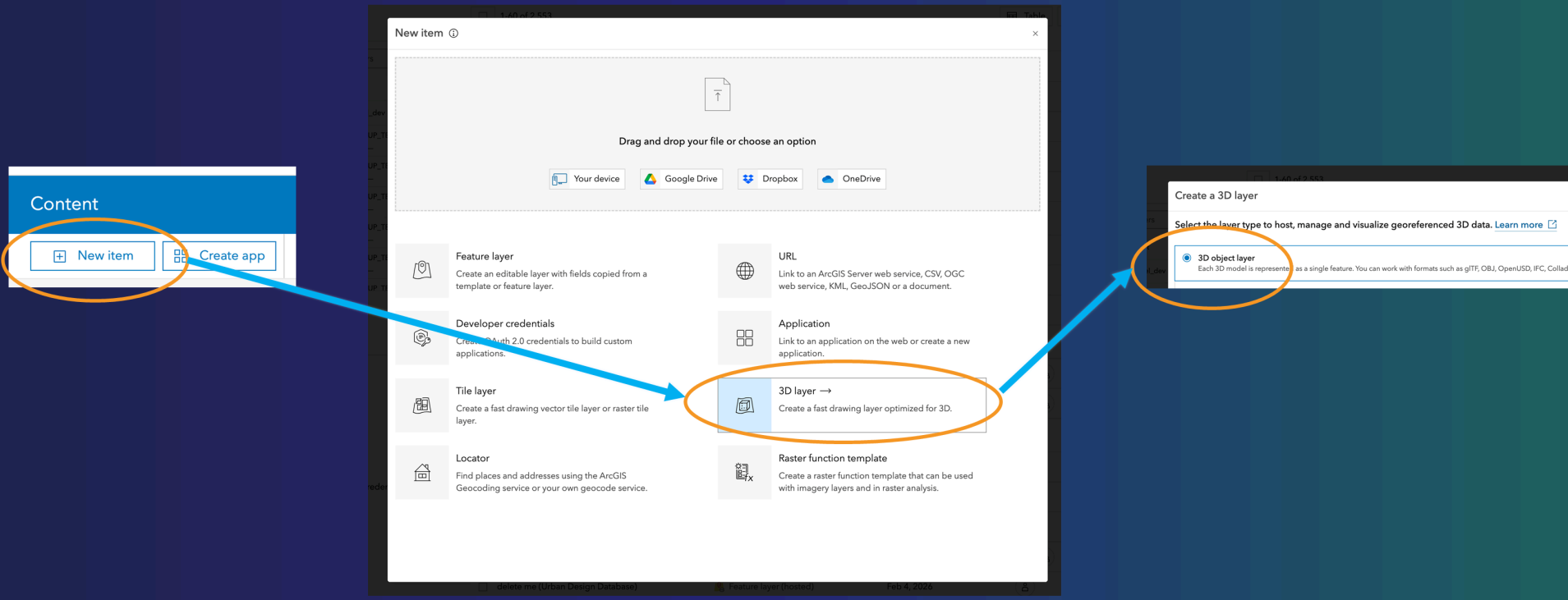
The SDK manages the connection.

> One 3D object layer combines optimized display and editable model records.

Creating 3D Object Layers

Creating 3D Object Layers

Portal



CREATE THE LAYER · BEHIND THE SCENES

Create the service, publish it, initialize the cache

1. Create the feature service
2. Publish the associated scene layer
3. Initialize the scene cache

CREATE PROGRAMMATICALLY

Create the layer programmatically

1 Create the feature service

`:portalUrl/content/users/:username/createService`

2 Publish the scene layer

`:portalUrl/content/users/:username/publish`

3 Rebuild the scene cache

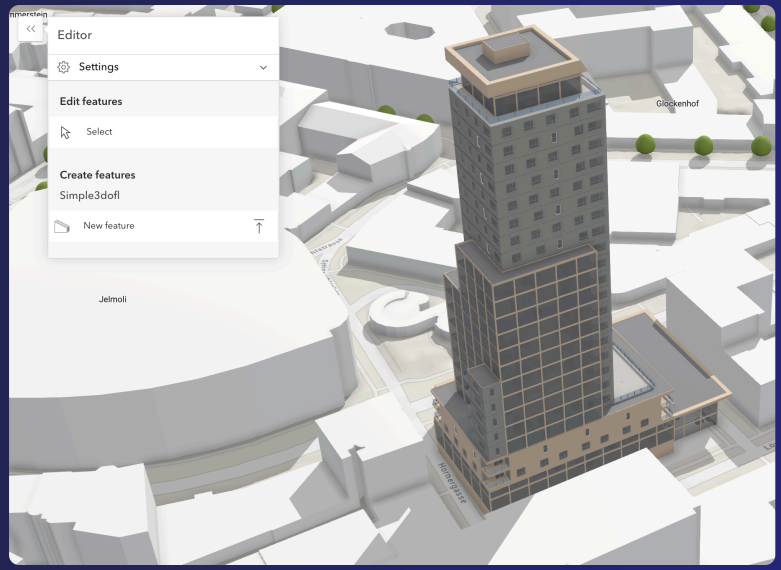
`:sceneServiceAdminUrl/rebuildCache`

```
params: {
  createParameters: {
    name,
    spatialReference: { wkid },
    serviceDataType: 'Object3D',
    capabilities,
    layers: [
      {
        fields,
        infoFor3D: {
          queryFormats: ['3D_shapebuffer', '3D_glb'],
        },
        hasZ: true,
      },
    ],
  },
  targetType: 'featureService',
  async: true,
```

Model upload and conversion

UPLOAD & CONVERSION

Use the Editor component for standard editing

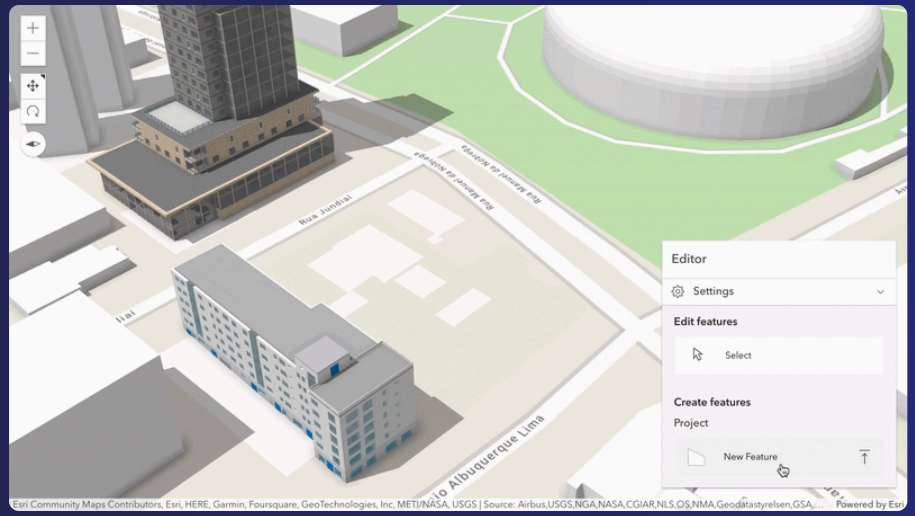



```
<arcgis-scene item-id="…">
  <arcgis-editor slot="top-right">
  </arcgis-editor>
</arcgis-scene>
```

Upload, place and edit models in your application.

CUSTOM WORKFLOWS

Use Mesh when you need control over the workflow



1. Convert the model.
2. Place and manipulate it.
3. Save the feature.

Open the SDK sample ↗

CONVERSION ≠ PERSISTENCE

Convert the model, then persist the feature

```
const { mesh } = await sceneLayer.convertMesh([file], {
  defaultOrigin: view.center.clone(),
});

await sceneLayer.applyEdits({
  addFeatures: [new Graphic({ geometry: mesh })],
});
```

> convertMesh → place / adjust → applyEdits

GEOREFERENCING

Give the model an origin in the real world

Embedded placement

Use the model's georeferencing information when available.

Fallback placement

Use `defaultOrigin` when the model has no intrinsic origin.

Explicit placement

Use `origin` to override intrinsic placement.

PRECISE PLACEMENT

Let the user enter exact coordinates

```
sketchViewModel.tooltipOptions.enabled = true;

sketchViewModel.place(mesh);
```

Use the placement tools for position, rotation and scale.

RESPOND TO MODEL METADATA

Check placement information before saving

```
const { mesh, georeferenceInfo } = await sceneLayer.convertMesh([file]);

if (georeferenceInfo) {
  const graphic = new Graphic({ geometry: mesh, symbol: meshSymbol });
  graphicsLayer.add(graphic);
  sketchViewModel.update(graphic);
  view.goTo(graphic);
} else {
  sketchViewModel.place(mesh);
}
```

> If the origin was reprojected, check the placement accuracy.

Vertex spaces

Two meshes. The same shape.
Two ways to locate their vertices.

Red · world coordinates | Blue · local offsets

Every vertex has longitude, latitude and elevation.

Hover or focus a number to show its point. Click to keep it highlighted.

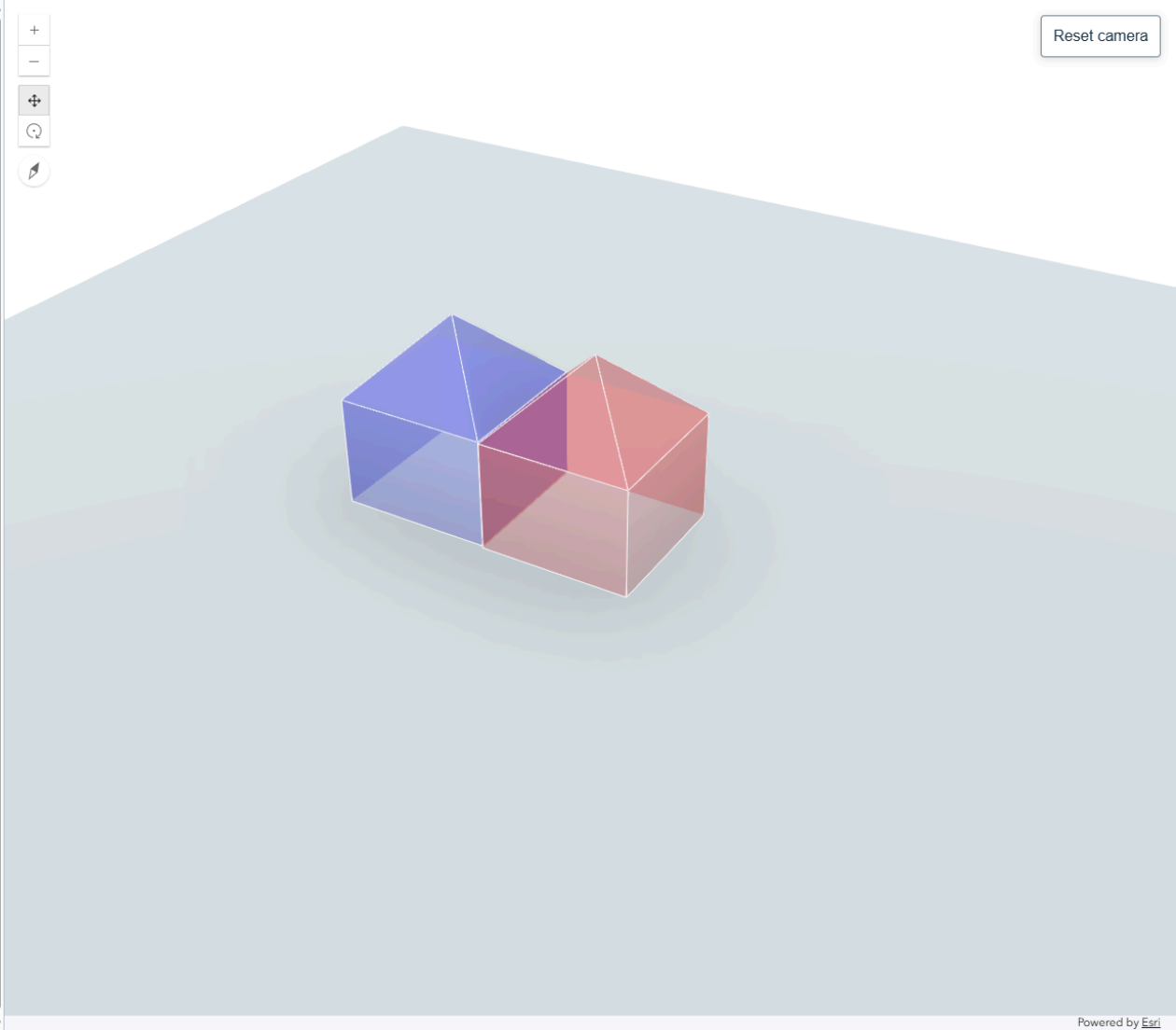
```
vertexSpace: new MeshGeoreferencedVertexSpace()

// x, y, z (rounded for display)
position: [
  -119.26676937, 35.23146354, 92.613, // 0
  -119.26672445, 35.23146354, 92.613, // 1
  -119.26672445, 35.23150022, 92.613, // 2
  -119.26676937, 35.23150022, 92.613, // 3
  -119.26674691, 35.23148188, 94.613, // 4
  -119.26676937, 35.23146354, 89.613, // 5
  -119.26672445, 35.23146354, 89.613, // 6
  -119.26672445, 35.23150022, 89.613, // 7
  -119.26676937, 35.23150022, 89.613, // 8
]
```

Faces connect vertices into triangles in both meshes.

```
// Vertex indices start at 0
faces: [
  0, 1, 4, // triangle 1
  1, 2, 4, // triangle 2
  // …
]
```

Scene code Mesh data

SDK 5.1.27



CONVERT BETWEEN REPRESENTATIONS

Convert vertex space for the operation you need

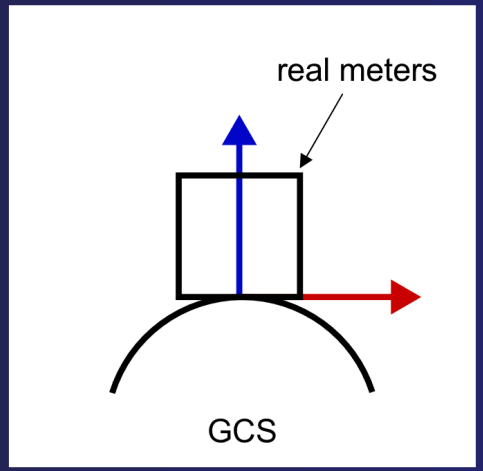
```
const origin = [center.x, center.y, center.z];

const localMesh = await convertVertexSpace(
  mesh,
  new MeshLocalVertexSpace({ origin }),
);
```

> A vertex offset becomes a map location through its origin and vertex space.

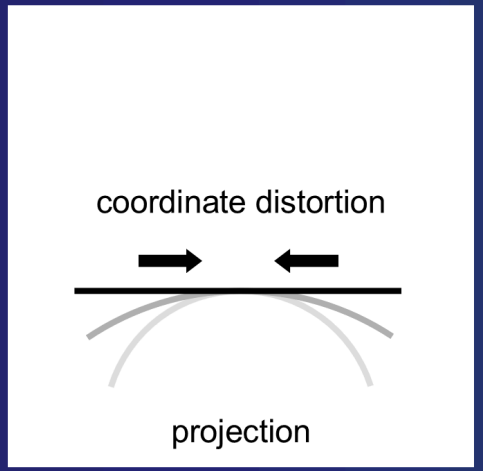
Spatial references

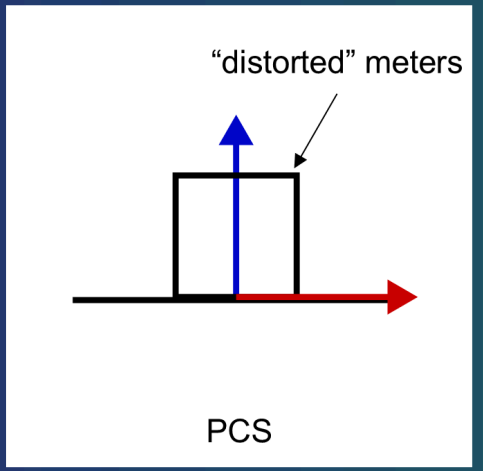
SPATIAL REFERENCES

Match the coordinate system to the scene



MeshLocalVertexSpace

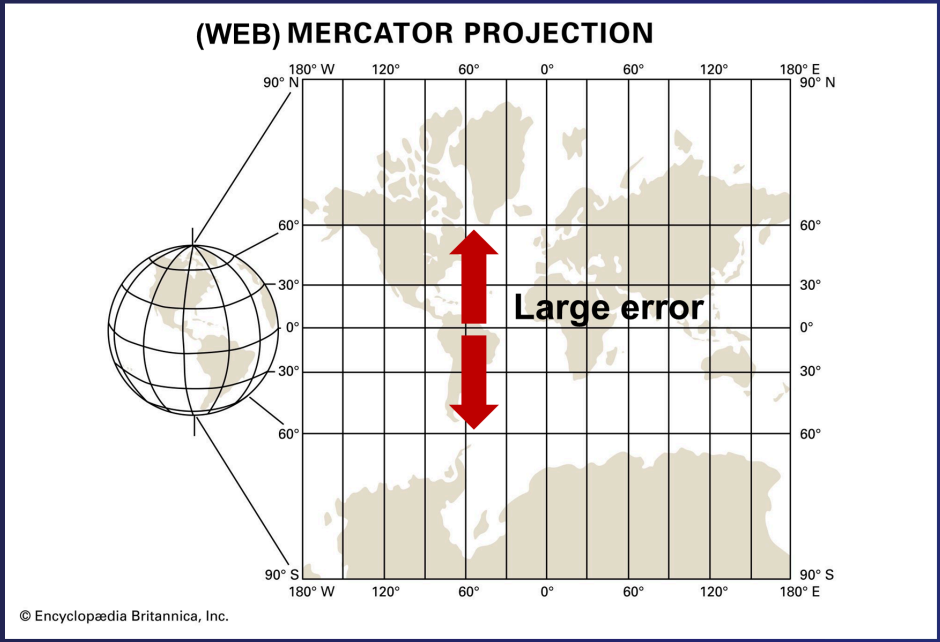




MeshGeoreferencedVertexSpace

WEB MERCATOR

Map distortion also affects 3D model placement



Scale distortion increases away from the equator.

> For a global or Web Mercator scene, use WGS84 for the 3D object layer.

CHOOSE BEFORE YOU PUBLISH

Which spatial reference should the layer use?

Is your project data in a local coordinate system? (e.g NAD83, LV95+)

yes

Use the same coordinate system for the 3D Object Layer

no

Use WGS84 for the 3D Object Layer

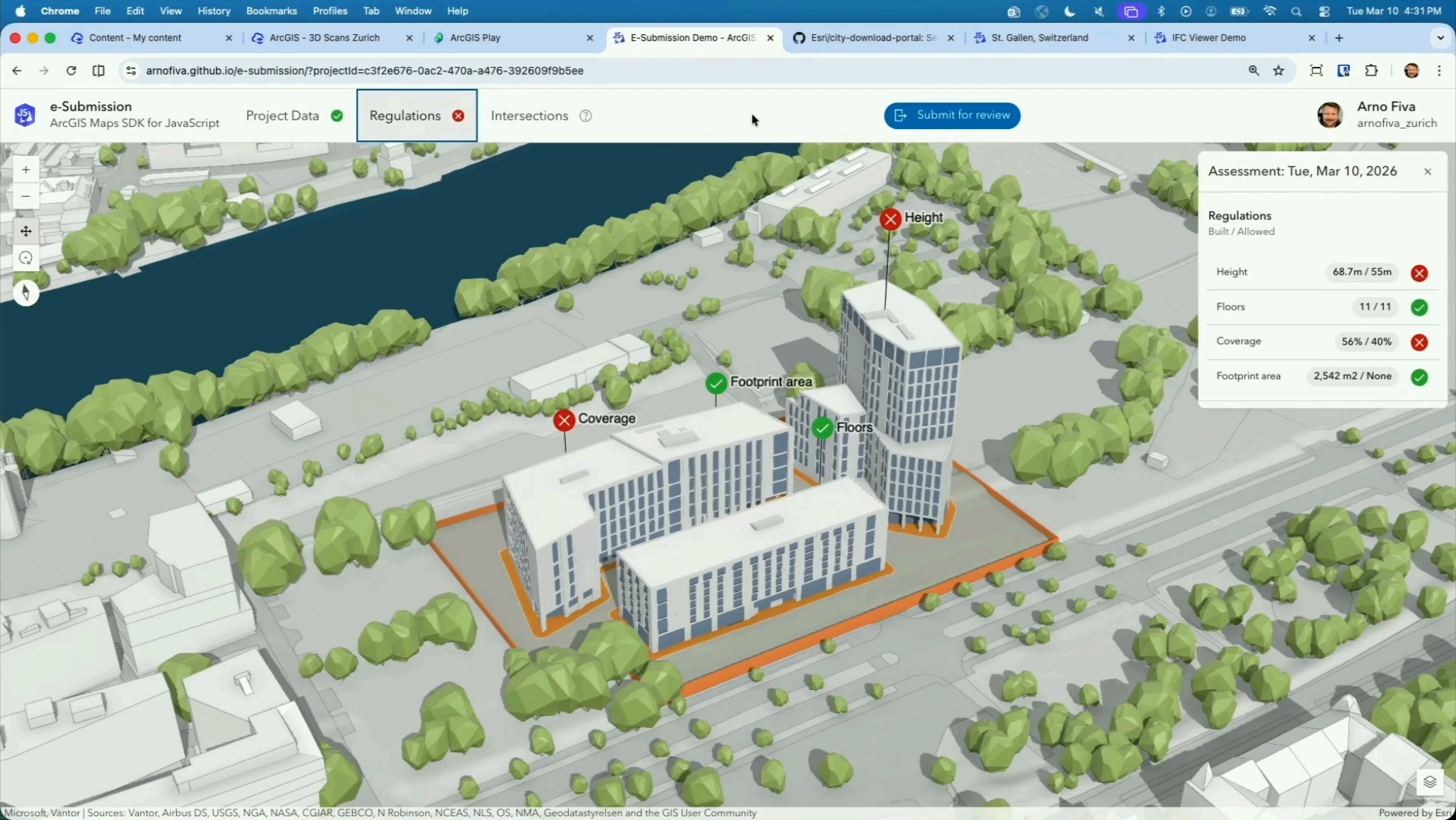
Query and manipulation

QUERY & MANIPULATION

Query a mesh—or just its footprint

```
const query = sceneLayer.createQuery();
query.objectIds = [objectId];
query.returnGeometry = true;

const { features } = await sceneLayer.queryFeatures(query);
const mesh = features[0].geometry;

await mesh.load();
```

`query.multipatchOption = 'xyFootprint'` requests the 2D footprint instead.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ArcGIS Play
E-Submission Demo - ArcGIS
Esri/city-download-portal: Se
St. Gallen, Switzerland
IFC Viewer Demo
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e-Submission
ArcGIS Maps SDK for JavaScript
Project Data
Regulations
Intersections
Submit for review
Arno Fiva
arnofiva_zurich
Height
Footprint area
Coverage
Floors
Assessment: Tue, Mar 10, 2026
Regulations
Built / Allowed
Height 68.7m / 55m
Floors 11 / 11
Coverage 56% / 40%
Footprint area 2,542 m2 / None
Microsoft, Vantor | Sources: Vantor, Airbus DS, USGS, NGA, NASA, CGIAR, GEBCO, N Robinson, NCEAS, NLS, OS, NMA, Geodatastyrelsen and the GIS User Community
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EXPORT

Export mesh geometry as a GLB model

```
await mesh.load();
const buffer = await mesh.toBinaryGLTF();

const blob = new Blob([buffer], {
  type: 'model/gltf-binary',
});
```

Building meshes + Terrain mesh → GLB → Modeling tools

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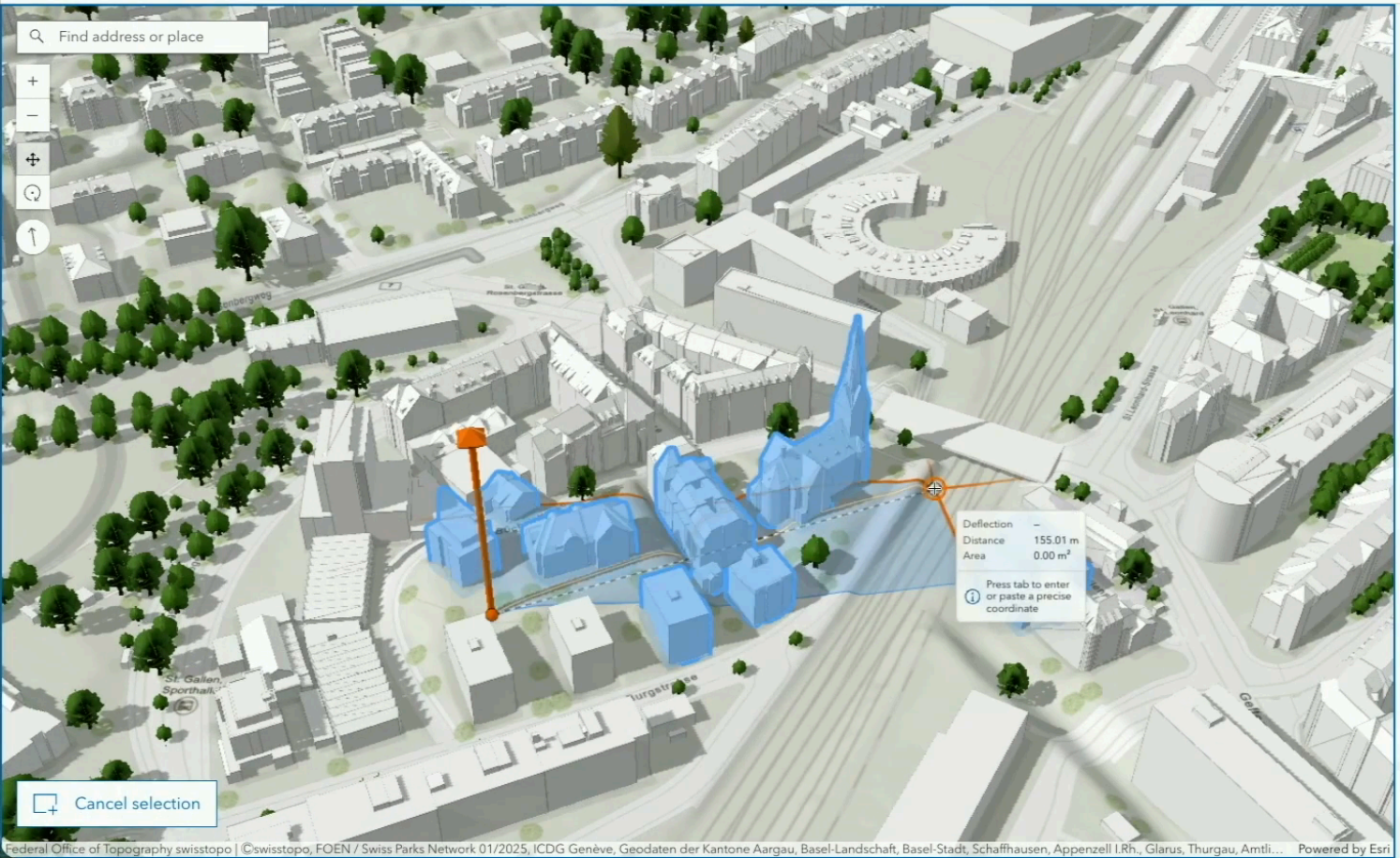
St. Gallen, Switzerland

St. Gallen is a Swiss city and the capital of the canton of St. Gallen. It evolved from the hermitage of Saint Gall, founded in the 7th century.



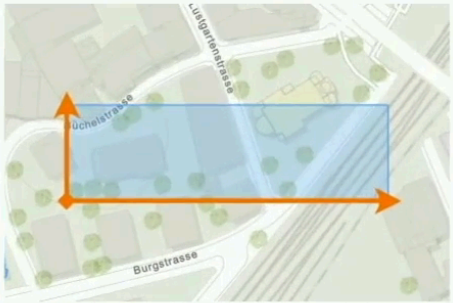
Model origin

x	Spatial reference (WKID)
2745250.72	CH1903+ / LV95 (2056)
y	**Elevation**
1253902.95	682.1 m

Set model origin

Copy to clipboard

Selection



North to south length
44.31 m

East to west length
148.54 m

Area
6,581.51 m²

Selected features
15

Update selection

Export

MODEL METADATA · SDK 5.1

Inspect source and derived models

```
const modelsByObjectId = await sceneLayer.queryModels({
  objectIds: [objectId],
});

const models = modelsByObjectId.get(objectId) ?? [];
for (const model of models) {
  console.log(model.format, model.isSource, model.size);
}
```

> One feature can have a source model and derived formats.
> Inspect metadata before downloading files.

DOWNLOAD ON DEMAND · SDK 5.1

Download the model the workflow needs

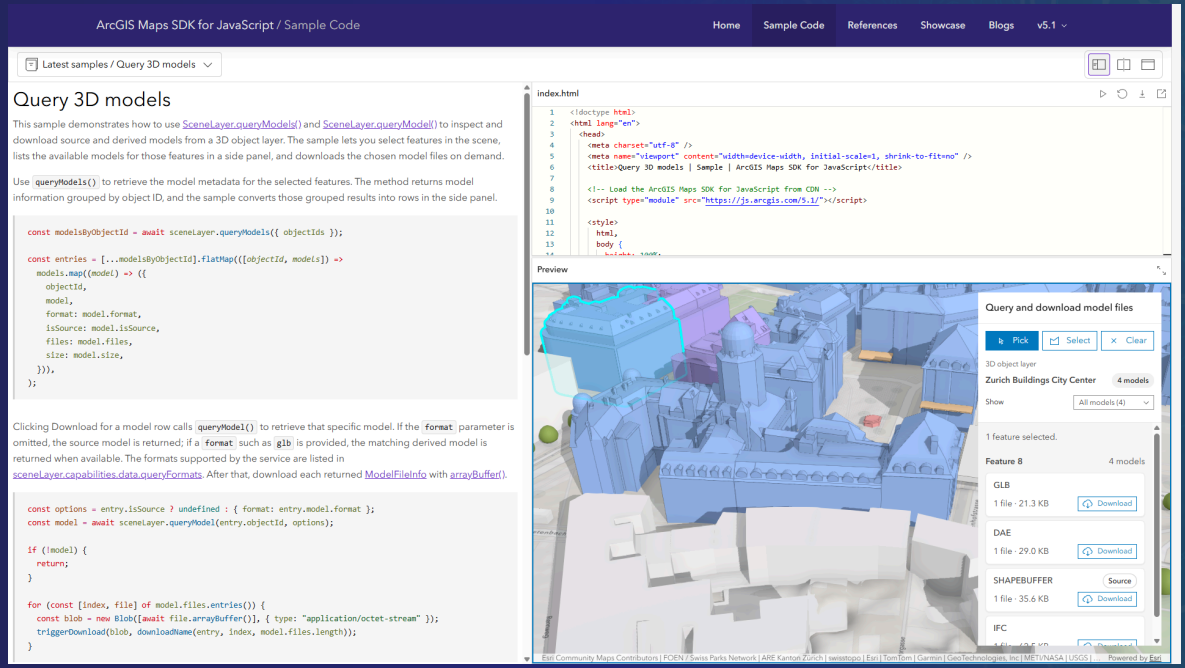
```
const source = await sceneLayer.queryModel(objectId);

const glb = await sceneLayer.queryModel(objectId, { format: 'glb' });

if (glb) {
  for (const file of glb.files) {
    const bytes = await file.arrayBuffer();
    // Offer these bytes as a file download.
  }
}
```

EXAMPLE · QUERY 3D MODELS

Select a building and download its models



Pick → inspect formats → download

Open sample and code ↗

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Display
Shell
Interior
Spaces
Levels
Level 5
Level 4
Level 3
Level 2
Level 1
Ground
Ground Floor
Level -1
IFC Model
Spaces
Area
Office
Residential
Commercial
Common Space
Other
IFC Model
Uploaded Projects
Built area
2433.17m2
Usable area (GFA)
2022.90m2
Office
Residential
Commercial
Common Space
Other
Esri Community Maps Contributors, FOEN / Swiss Parks Network, ARE Kanton Zürich, swisstopo, Esri, TomTom, Garmin, GeoTechnologies, Inc, METI/NASA, USGS | Sources: Vantor, Airbus DS, USGS, NGA, NASA, CGIAR, GEBCO, N Robins...
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Takeaways

API RECAP

Three APIs carry the core workflow

SceneLayer

`convertMesh()` · `applyEdits()`

`queryFeatures()`

`queryModels()` · `queryModel()`

Mesh

`vertexSpace`

`transform`

`toBinaryGLTF()`

meshUtils

`convertVertexSpace()`

Convert coordinates for the next operation.

TAKEAWAYS

Store the model. Locate it correctly. Build on it.

1

Use the layer as the record

Keep original assets and manage features through the SDK.

2

Choose coordinates deliberately

Align spatial references, origins and vertex spaces.

3

Extend the workflow

Analyze geometry, export models or interpret source data.

RESOURCES

Continue with the guides and working examples

Learn the workflow

- 3D object workflows in the SDK
- Upload models and applyEdits sample
- 3D Object Layer overview
- SceneLayer reference

Explore the examples

- City Download Portal — source
- Custom IFC viewer — source
- Query source and derived models — sample
- Original Esri session recording
- Vertex spaces — live code comparison

Questions

Which 3D workflow are you building?

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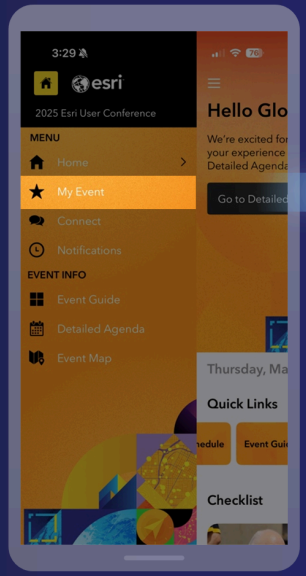

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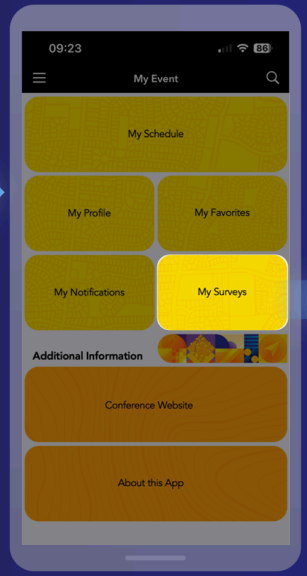

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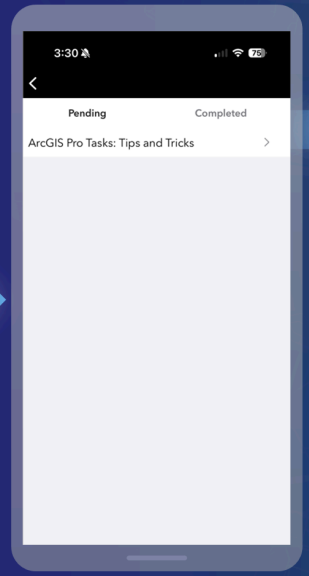

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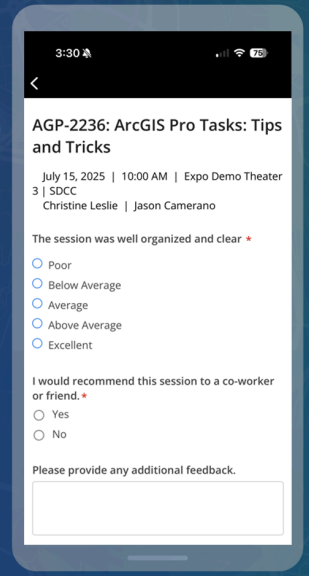

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